

AUTO-MASTER ERP & WORKSHOP ECOSYSTEM

Comprehensive Specification: Garage Operations, Spare Parts POS, Yard Storage ('Field Money'), Tool Hire & Master Admin Control

OPERATOR MODEL:

Single Owner / Central Control

PAYMENT CHANNELS:

M-Pesa (Till/Paybill) + Cash + Split

DOCUMENT PURPOSE:

Complete Technical & Business Spec

1. Executive Overview & System Architecture

This system addresses the exact operational realities of a full-service automotive center managed under **one person/owner**. It integrates counter spare-parts retail, vehicle bodywork/fabrication job cards, yard parking monetization, tool rentals, and strict shift/stock auditing into a single real-time ledger.

Unified Operational Pipeline:

Spare Parts Store → Garage Job Cards (Labor + Store Parts + Outside Purchases) → Yard Parking Fee Tracker (KES 5,000/week) → Tool Rental Desk → Shift Audit & Stock Taking → Master Admin Dashboard (Real-time P&L;)

2. Module 1: Spare Parts Inventory & POS (Counter Sales)

- Unique Item Tracking:** Every item is tracked with SKU, description, and unique Serial/IMEI numbers where applicable (e.g., diagnostic kits, specialized electronics, batteries).
- Cost vs. Selling Margin:** Buying price, Selling price, and Live Gross Profit margin calculated on every sale.
- Stock Decrement & Low-Stock Alerts:** Real-time inventory deduction upon checkout with automatic low-stock notifications.
- Fast Multi-Payment POS:** Supports Cash, M-Pesa (Till/Paybill confirmation), and Split billing (e.g., KES 2,000 Cash + KES 3,500 M-Pesa).
- Stock Movement Equation:** Closing Stock = Opening Stock + New Stock In - Counter Sales - Garage Job Usage.

3. Module 2: Garage Workshop & Job Cards (Repairs & Builds)

For vehicles built from scratch, undergoing major body restoration, or receiving routine maintenance.

Labor & Workmanship Billing

- Itemized breakdown for **Panel Beating, Welding, Spray Painting, Mechanical, and Electrical**.
- Assigned mechanics logged per task with stage tracking (Intake → In Progress → Completed).
- Formal quotations generated for client approval before work starts.

Parts & Outside Procurement

- In-House Parts:** Deducted from workshop stock at selling price.
- Outsourced Purchases ('Bought Outside'):** Expense logged with external vendor receipts, charged to vehicle invoice with optional markup.
- Completion Settlement:** Deposit deducted, balance paid via M-Pesa/Cash.

Job Card Invoice Formula:

Total Bill = [Labor (Panel + Welding + Paint)] + [Store Parts] + [Outside Purchases (+ Markup)] - [Advance Deposit]

4. Module 3: Yard, Parking & 'Field Money' Tracker

Monetizes vehicles parked in the yard, holding area, or awaiting customer pickup/clearance.

- **Weekly Field Fee:** Standard automated accrual (e.g., KES 5,000 / week per bay).
- **Overstay & Storage Penalties:** Automatic daily/weekly penalty surcharge if vehicle exceeds authorized time.
- **Combined Vehicle Invoicing:** If a car had both repair jobs and yard parking, the bill consolidates seamlessly.
- **Digital Gate Clearance Pass:** Barcode / SMS gate pass produced only when the account balance reaches zero.

5. Module 4: Tool & Equipment Rental Desk

Tracks rental income and protects high-value workshop machinery hired out to third parties or freelance mechanics.

Tool / Machine	Identifier	Hire Rate	Security Deposit	Audit & Condition Rules
Body Jack	BJ-01 / BJ-02	KES 1,000 / Day	KES 3,000	Logged with Hirer ID, Phone & Return Timer
Check Handle / Lever	CH-10	KES 500 / Day	KES 1,500	Pre-rental & post-return damage inspection
Welding Machine	WM-04	KES 2,000 / Day	KES 5,000	Electrode / clamp check + runtime log
Hydraulic Puller	HP-03	KES 1,500 / Day	KES 4,000	Hydraulic seal and ram pressure check

6. Module 5: Shift Management & Physical Stock Taking

- **Shift Handover & Drawer Balancing:** Cashier inputs opening float; system tallies cash and M-Pesa collected; flags any cash drawer shortages.
- **Physical Stock Taking:** Compares physical count vs. system recorded quantity. Computes exact monetary shrinkage/theft loss.
- **Audit Trail:** Every inventory modification, outside purchase, and discount requires an authenticated log entry.

7. Master Admin Dashboard (Single Owner Control Center)

The central command center giving the single business owner real-time oversight over all operations:

Dashboard Section	Real-Time KPI & Intelligence Displayed	Owner Action Trigger
Financial Pulse	Today's Gross Sales, Monthly Net P&L;, M-Pesa Total, Cash Drawer Total, Total Stock Valuation	Filter date range, export Excel/PDF, reallocate funds
Garage Job Board	Active Cars, Panel/Welding Stage, Outside Purchases, Invoiced Total, Deposits, Pending Balances	1-Click Approve Quotation, Add Outside Receipt, Settle Job
Yard & Field Matrix	Vehicles on field, Accumulated 5k/wk fees, Overstay alerts, Total parking balance	Issue Gate Pass, apply storage waiver or penalty
Inventory & POS	Top selling parts, Low stock warnings, Profit margin per item, Stock movement	Adjust prices, log batch shipment, trigger reorder
Tool Rental Desk	Active rentals (Body jacks, handles), Due dates, Held deposits, Overdue penalties	1-Click Check-in, refund deposit, charge damage fee
Shift & Audit Hub	Cashier shift status, Cash variance / shortage alerts, Stock taking audit variances	Lock shift, review cashier handover, approve adjustments